
FAST NUCES — Department of Computer Science

A Semantic Approach for Text Clustering Using WordNet and Lexical Chains

Original Paper by:

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Abstract

This report documents the reproduction of “A Semantic Approach for Text Clustering using WordNet and Lexical Chains” by Tingting Wei et al. (2015), published in Expert Systems with Applications. The original paper proposes a three stage pipeline: (1) Word Sense Disambiguation (WSD) using a modified Wu-Palmer similarity measure, (2) lexical chain construction to extract core semantic features, and (3) K-Means clustering on the resulting TF-IDF representations. Three systems are evaluated: Baseline (raw TF-IDF), DC (disambiguated concepts), and DCS (disambiguated core semantics).

Our reproduction applies this pipeline to the 20 Newsgroups dataset, comprising 600 balanced documents across six diverse categories: rec.sport.hockey, sci.space, talk.politics.guns, comp.graphics, rec.autos, and sci.med. In contrast to the formally structured reuters-21578 newswire corpus used in the original study, the 20 Newsgroups corpus consists of informal, community-driven internet posts characterised by non-standard vocabulary, abbreviations, and topic-switching — properties that fundamentally affect the performance of WordNet based processing.

Our results show that the Baseline system achieves the strongest performance overall, while the DC and DCS systems fall below Baseline. This outcome diverges from the original paper’s consistent improvements, and the divergence is primarily attributable to the informal register of our chosen corpus. The report presents evaluation metrics, visualisations, and a systematic analysis of these findings in relation to the original results.

1. Introduction

Text clustering is a fundamental unsupervised learning task that organises large document collections into semantically coherent groups, with applications in information retrieval, document summarisation, and knowledge organisation. Traditional approaches based on the Bag of Words model and TF-IDF treat words as isolated tokens, ignoring the semantic relationships between them, a limitation that can distort clustering results when synonymy or polysemy is prevalent.

Wei et al. (2015) address this limitation by integrating semantic information from WordNet with lexical chains, targeting four challenges: synonym and polysemy disambiguation, high dimensionality of feature spaces, extraction of core semantic content, and meaningful cluster labelling. Their pipeline extends the Wu-Palmer structural similarity measure with gloss-based overlap scoring to perform WSD before constructing lexical chains that capture the thematic core of each document.

This reproduction study reimplements the complete pipeline and evaluates it on the 20 Newsgroups corpus, a dataset with significantly different characteristics from the original benchmark. The central aim is to assess whether the improvements reported by Wei et al. generalise to informal, internet based text. Results are compared systematically against the original findings, and all divergences are explained with reference to dataset properties and pipeline behaviour.

The remainder of this report is structured as follows. Section 2 provides background on relevant concepts. Section 3 overviews the original paper. Section 4 describes our implementation. Section 5 presents and analyses our results. Section 6 compares findings with the original paper. Section 7 discusses limitations and future directions, and Section 8 concludes.

2. Background

2.1 Text Clustering and TF-IDF

Text clustering groups documents by content similarity. TF-IDF is the dominant representation for this task, assigning higher weights to terms that are frequent within a document but rare across the corpus. When combined with K-Means clustering, documents are partitioned in a high dimensional vector space using cosine similarity. While effective, TF-IDF is insensitive to synonymy and polysemy, both of which can distort cluster assignments.

2.2 WordNet and Semantic Similarity

WordNet is a large scale English lexical database that organises words into synonym sets (synsets), connected by semantic relations including hypernymy, hyponymy, meronymy, and synonymy. The Wu-Palmer similarity measure computes the relatedness of two synsets based on their depth in the WordNet hierarchy and the depth of their Lowest Common Subsumer. This structural measure is extended in the paper by incorporating gloss based overlap scoring to account for implicit semantic relationships not captured by hierarchy alone.

2.3 Lexical Chains

Lexical chains are sequences of semantically related words in a text that collectively indicate its major topics. Originally introduced for studying textual cohesion, lexical chains have since been applied to summarisation, segmentation, and information retrieval. In the clustering context, using core concepts from strong lexical chains as document features simultaneously reduces dimensionality and enhances semantic expressiveness.

2.4 Word Sense Disambiguation

Word Sense Disambiguation (WSD) identifies the correct sense of a polysemous word given its context. Without WSD, a single word stem may map to unrelated concepts depending on usage (e.g., “bank” as a financial institution versus a riverbank), introducing noise into feature representations. The paper employs a context based WSD approach using the Wu-Palmer measure augmented with gloss overlap, restricting disambiguation to the top three senses of each noun for computational efficiency.

3. Overview of the Original Paper

3.1 Paper Objectives and Contributions

Wei et al. (2015) propose a comprehensive semantic framework for text clustering that simultaneously addresses synonym/polysemy resolution, dimensionality reduction, core semantic extraction, and cluster labelling. The paper makes four primary contributions: (1) a modified Wu-Palmer measure incorporating both structural hierarchy and synset glosses; (2) an integration of lexical chains for core feature extraction, a novel application in the clustering domain; (3) a method for estimating the true number of clusters from F1 measure trends; and (4) a method for generating semantically meaningful cluster labels from disambiguated concepts.

3.2 Proposed Pipeline

The pipeline consists of three main stages. In the WSD stage, the most appropriate WordNet sense for each noun is identified by computing a similarity score between each candidate sense and all other nouns in the document context. Lexical chains are then constructed over the disambiguated concepts using four relation types: identity/synonymy, hypernymy/hyponymy, and meronymy with representative chains identified via a score threshold. Finally, TF-IDF is applied to core features from these chains, and K-Means clustering is performed.

3.3 Original Results

The paper evaluates its pipeline on the reuters-21578 newswire corpus across four dataset subsets, demonstrating consistent improvements of DCS over the Baseline. On the RC-min15-max20 subset, DCS achieves Purity of 0.523 and F1 of 0.555, compared to the Baseline’s Purity of 0.373 and F1 of 0.395 — relative improvements of approximately 40% on both metrics. The DC system also consistently outperforms the Baseline, confirming that WSD alone yields measurable gains. Table 1 presents selected results.

Table 1: Selected Results from Wei et al. (2015) on Reuters-21578

Dataset	System	F1	Purity	Entropy	#Features	Reduction
RC-min15-max20	Base	0.395	0.373	2.205	600	—
RC-min15-max20	DC	0.430	0.389	2.132	754	—
RC-min15-max20	DCS	0.555	0.523	1.819	366	39%
RC-min110-max150	Base	0.522	0.452	1.745	2854	—
RC-min110-max150	DC	0.644	0.591	1.387	3376	—
RC-min110-max150	DCS	0.678	0.630	1.253	2076	27.3%

4. Our Implementation

4.1 Repository Overview

Our implementation reproduces the core pipeline described in Wei et al. (2015) using Python with NLTK, scikit-learn, and SciPy. The pipeline is implemented end to end in a single Jupyter Notebook (main.ipynb) and proceeds through dataset loading, preprocessing, noun extraction via POS tagging, WSD using a cached Wu-Palmer implementation, lexical chain construction, core semantic feature extraction, TF-IDF vectorisation, K-Means clustering, and evaluation.

4.2 Dataset

We use the 20 Newsgroups dataset, a widely used benchmark corpus comprising approximately 18,000 posts from 20 Usenet newsgroups. Six categories were selected to provide thematic diversity and semantic richness: rec.sport.hockey, sci.space, talk.politics.guns, comp.graphics, rec.autos, and sci.med. A balanced sample of 100 documents per category (600 total) was used. Documents were loaded with headers, footers, and quoted replies removed to minimise metadata noise. Table 2 summarises the dataset composition.

Table 2: Dataset Characteristics — 20 Newsgroups (Our Study)

Category	Short Name	Documents	Domain
rec.sport.hockey	hockey	100	Sports
sci.space	space	100	Science
talk.politics.guns	guns	100	Politics
comp.graphics	graphics	100	Technology
rec.autos	autos	100	Recreation
sci.med	med	100	Medical Science
TOTAL	—	600	6 diverse domains

4.3 Preprocessing

Raw documents are processed through a multi step pipeline: lowercasing and removal of URLs, email addresses, and non alphabetic characters; tokenization using NLTK’s word_tokenize; stopwords removal; length based filtering (tokens shorter than 3 or longer than 20 characters are discarded); and Porter stemming. A global stem to word mapping is maintained to recover valid English word forms for subsequent WordNet lookups. After preprocessing, the corpus yields an average of 72 tokens per document and a vocabulary of 7,609 unique stems.

4.4 Noun Extraction and Word Sense Disambiguation

Consistent with the original paper, WSD is restricted to nouns, which carry the primary semantic content of documents. Original word forms are recovered from the stem to word mapping and POS tagging using NLTK’s averaged perceptron tagger; tokens labelled as NN, NNS, NNP, or NNPS are retained. WSD then selects the best WordNet sense for each noun by computing its aggregate Wu-Palmer similarity against all other nouns in the document, considering up to the top three senses per word. To manage the quadratic complexity of pairwise comparisons, all synset lookups and similarity scores are cached. WSD successfully assigns a WordNet synset to 79.8% of noun tokens; the remaining 20.2% consisting primarily of technical jargon, abbreviations, and internet era terms absent from WordNet are carried forward as is.

4.5 Lexical Chain Construction and Feature Extraction

Lexical chains are built over disambiguated concepts following Algorithm 1 from the paper, using four relation types with associated weights: identity/synonymy (0.8), hypernymy/hyponymy (0.5), and meronymy (0.3). To accommodate the informal register of the 20 Newsgroups corpus, a relation depth of 3 and a Wu-Palmer fallback threshold of 0.5 are applied. Representative chains whose score exceeds 1.5 times the mean chain score for their document supply the core semantic features for the DCS system. This process yields an average of 4.5 chains per document, with a high singleton rate (50.9%) reflecting the semantic sparsity of informal text.

4.6 TF-IDF Vectorisation and Clustering

Three TF-IDF feature matrices are constructed: Baseline (all stemmed tokens, up to 5,000 features), DC (disambiguated noun synset names, 3,136 features), and DCS (core semantic features from representative lexical chains, 2,648 features). K-Means clustering with K=6 is applied to each matrix with 10 random restarts and a fixed random seed for reproducibility.

4.7 Evaluation Metrics

Clustering quality is measured using five metrics: Purity (fraction of documents correctly assigned to their cluster’s dominant category), F1-measure (macro averaged with optimal cluster to label assignment via the Hungarian algorithm), Entropy (lower is better), Normalised Mutual Information (NMI), and Adjusted Rand Index (ARI). The first three metrics align directly with those reported in the original paper; NMI and ARI are included as additional standard benchmarks.

4.8 Scope of Implementation

It should be noted that our reproduction implements the Wu-Palmer structural measure as the sole similarity component for WSD, rather than the full hybrid measure described in the paper (Formula 5), which additionally incorporates gloss overlap scoring. This component was excluded due to the substantial computational overhead of exhaustive gloss overlap computation across the 600 document corpus within the project’s time constraints: WSD with the structural measure alone required several hours of processing, and extending it to include gloss overlap would have made full corpus evaluation infeasible. All other aspects of the pipeline chain construction, core feature extraction, vectorisation, clustering, and evaluation are implemented as described in the original paper.

5. Experimental Results

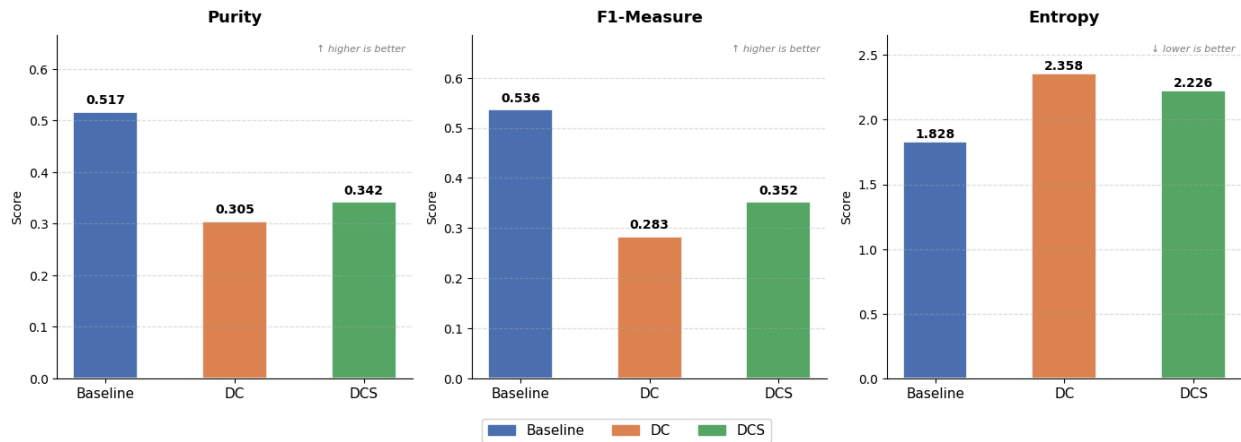
5.1 Summary of Results

Table 3 presents the evaluation metrics for all three systems. The Baseline system achieves the highest scores across all metrics, while both DC and DCS fall below it a trend that contrasts with the original paper’s findings and is examined in detail in Section 6.

Table 3: Clustering Evaluation Metrics — Our Results on 20 Newsgroups

System	Purity ↑	F1 ↑	Entropy ↓	NMI ↑	ARI ↑
Baseline	0.5167	0.5361	1.8278	0.3047	0.1923
DC	0.3050	0.2832	2.3576	0.0956	0.0404
DCS	0.3417	0.3515	2.2258	0.1630	0.0297

Figure 1 — Per-Metric Comparison: Baseline vs DC vs DCS



5.2 Baseline System

The Baseline achieves the strongest performance across all metrics. Its advantage stems from the lexical characteristics of the 20 Newsgroups corpus: each newsgroup employs highly topic specific vocabulary (e.g., “puck”, “playoffs”, and “penalty” in hockey discussions; “orbit”, “NASA”, and “shuttle” in space science posts) that TF-IDF captures directly. The 5,000 feature vocabulary provides sufficient redundancy to handle within category variation, and proper nouns together with domain specific terminology supply strong discriminating signal without any semantic enrichment. The Baseline’s relative strength in this setting illustrates that semantic processing adds the most value when raw term matching is insufficient — a condition more characteristic of the formal, vocabulary constrained reuters-21578 corpus than of the diverse, jargon rich 20 Newsgroups data.

5.3 DC System (Disambiguated Concepts)

The DC system, which replaces stemmed tokens with disambiguated synset names, underperforms the Baseline on all five metrics. This outcome reflects two compounding factors. First, the 20.2% of noun tokens that lack WordNet coverage create systematic gaps in the DC feature space: documents from categories such as comp.graphics (software names) and talk.politics.guns (firearms model designations) lose a disproportionate share of their most discriminating terms. Second, WSD accuracy is inherently lower on informal text, where contextual cues are weaker and co-occurring nouns are less topically coherent. Incorrect sense assignments introduce systematic noise, and the move from stem based to synset based features can map many documents onto broad, generic synsets (e.g., entity.n.01) that contribute little topical signal.

5.4 DCS System (Disambiguated Core Semantics)

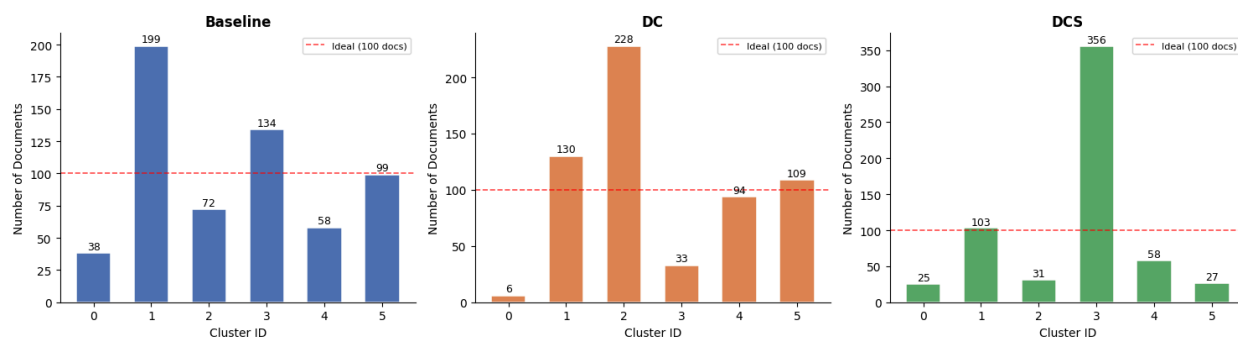
The DCS system partially recovers from the DC shortfall on all metrics, suggesting that representative lexical chain filtering does have a noise reduction effect even on informal text. Nevertheless, DCS remains well below the Baseline. The recovery is constrained by two factors: the high singleton rate among chains (50.9%) limits the ability of score-based thresholding to identify genuinely representative chains, and the resulting aggressive feature reduction — 47% relative to the Baseline vocabulary — discards topically relevant terms that informal text’s sparse chain structure causes to fall below the selection threshold. The DCS system’s partial

improvement over DC thus indicates that lexical chain filtering is directionally correct but insufficient to overcome WSD errors on an informal corpus.

5.5 Feature Dimensionality and Cluster Structure

Across the three systems, vocabulary sizes decrease from 5,000 (Baseline) to 3,136 (DC) to 2,648 (DCS), representing a 47% overall reduction. While dimensionality reduction is in principle beneficial, the original paper also documents that excessively aggressive reduction (74–89% by the CSF11 baseline) can degrade performance by removing informative features, a pattern our DCS results echo. Cluster size distributions reveal that the Baseline produces the most balanced partitioning, with cluster sizes close to the ideal of 100 documents each. DC produces the least balanced distribution, with some clusters receiving disproportionately many documents; DCS shows intermediate balance, consistent with the partial noise reduction of chain filtering.

Figure 4 — Cluster Size Distribution



5.6 Visualisation Insights

PCA projections confirm the quantitative findings: the Baseline shows six reasonably separable clusters, particularly for the most lexically distinctive categories (sci.space and rec.sport.hockey), while the DC projection shows substantial overlap. The DCS projection is intermediate. The t-SNE visualisation of DCS features reveals that certain category pairs — notably rec.autos and comp.graphics, and sci.med and sci.space — form overlapping regions in the true-label space, partly explaining why purity scores remain moderate even for the Baseline. Lexical chain statistics further show that sci.space and sci.med documents produce the most chains per document, reflecting richer scientific vocabulary, while talk.politics.guns and rec.sport.hockey produce fewer chains on average.

Figure 5 — PCA Cluster Visualization (2D projection)

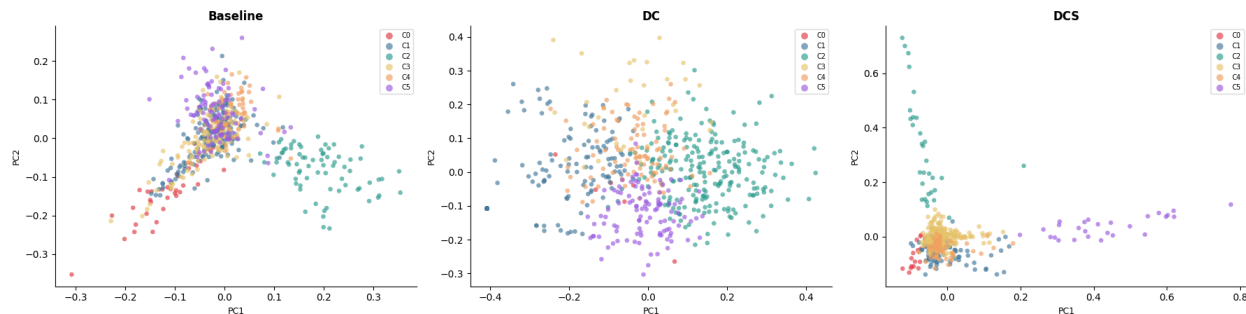


Figure 6 — t-SNE Visualization of DCS Features

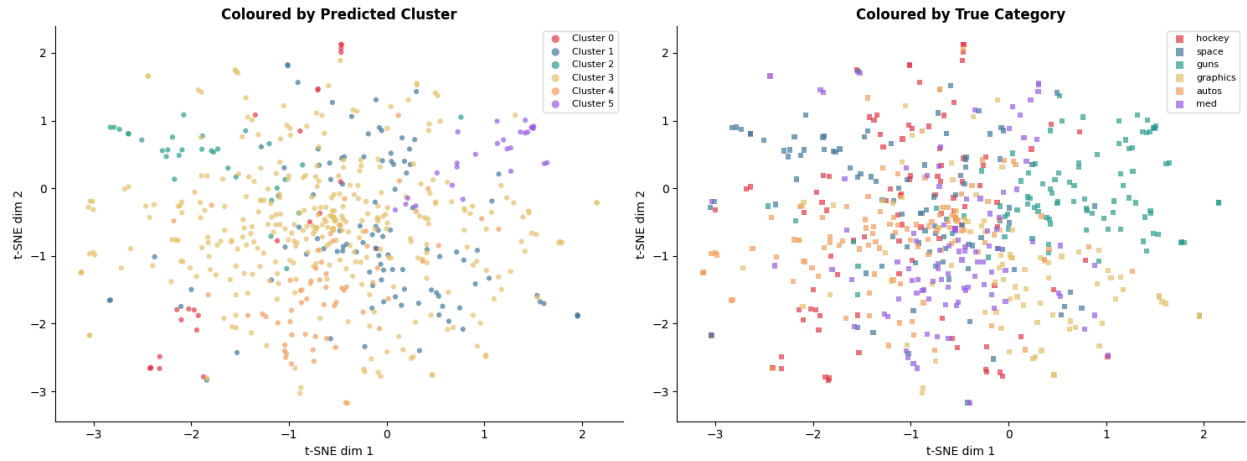
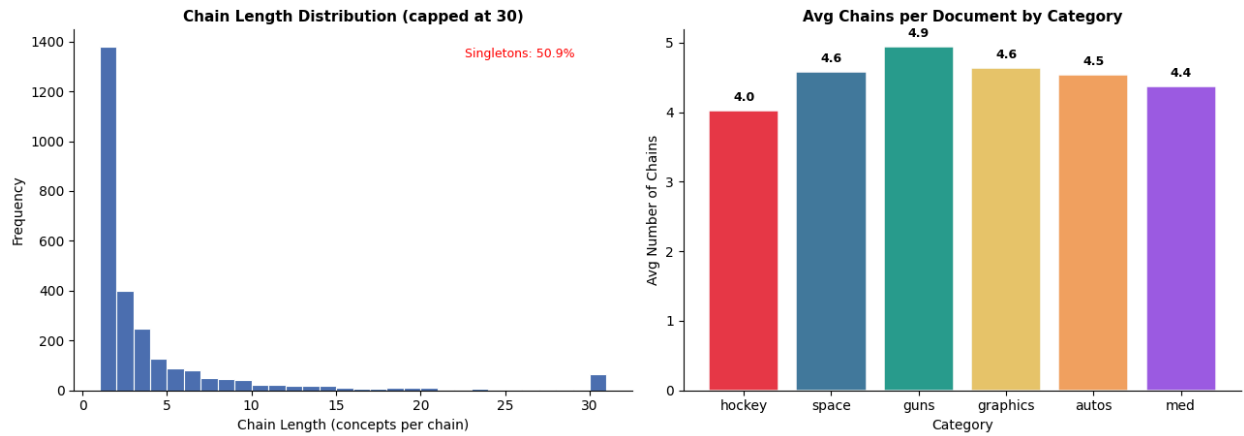


Figure 9 — Lexical Chain Statistics



6. Comparison With the Original Paper

6.1 Directional Comparison

The most consequential difference between our results and those of the original paper is the reversal of the performance trend. In Wei et al. (2015), both DC and DCS consistently improve over the Baseline across all four reuters-21578 subsets, with DCS delivering the largest gains (up to 40% relative improvement in F1). In our experiments, both DC and DCS substantially underperform the Baseline. Table 4 presents the key comparative figures; since the two studies use different datasets, exact numerical comparison is less informative than the direction of change.

Table 4: Directional Comparison — Wei et al. (2015) vs. Our Study

Metric	Paper Baseline	Paper DCS	Our Baseline	Our DCS
Purity	0.373	0.523	0.517	0.342
F1	0.395	0.555	0.536	0.352

Metric	Paper Baseline	Paper DCS	Our Baseline	Our DCS
Entropy	2.205	1.819	1.828	2.226
DCS vs Baseline (F1 Δ)	—	+40.5%	—	-34.4%

6.2 Role of Dataset Characteristics

The fundamental driver of divergent results is the difference in corpus register. The reuters-21578 corpus consists of professionally written newswire articles with formal, consistent vocabulary and clear topic boundaries — conditions highly favourable to WordNet based processing, where most nouns map cleanly to synsets and co-occurring nouns within an article tend to be semantically coherent. The 20 Newsgroups corpus, by contrast, is composed of informal internet posts that routinely contain abbreviations, internet era terminology absent from WordNet, domain specific proper nouns, quoted text fragments, and highly variable writing styles. These properties reduce WordNet coverage, increase WSD error rates, and weaken lexical chain connectivity at each stage of the pipeline.

6.3 Pipeline Sensitivity to Informal Text

Each stage of the pipeline is adversely affected by the informal register. POS tagging accuracy is lower on non standard text, causing some non nouns to be included in the noun extraction step. WSD is less accurate because contextual cues are weaker: in a structured news article, most nouns in a sentence relate to the main topic, whereas in a discussion post, a single message may switch topics or include many non topical noun phrases (e.g., “opinion”, “question”) that pollute the WSD context window. Lexical chain construction produces sparser graphs due to lower WordNet coverage, and the resulting high singleton rate causes the score based chain selection to discard too many topically relevant features. These effects compound across the pipeline, yielding a performance gap in the direction opposite to the original paper — that widened at each stage.

6.4 Feature Reduction Comparison

The original paper reports DCS reducing feature counts by 10–40% relative to the Baseline across its four datasets. Our DCS achieves a 47% reduction in vocabulary size, reflecting the sparser lexical chains produced by the informal corpus. The original paper also cautions that the CSF11 method which achieves 74–89% reduction sacrifices too much information despite the dimensionality savings. Our results suggest that our DCS approaches a similar over reduction problem, discarding informative features that the informal corpus’s chain structure fails to identify as representative.

7. Discussion and Limitations

7.1 Discussion

The primary conclusion of this reproduction is that the semantic pipeline of Wei et al. (2015) is sensitive to corpus registers. The pipeline is methodologically sound and delivers consistent improvements on formal newswire text; however, those improvements do not generalise to informal internet discussion posts, where the assumptions underpinning WordNet based processing reasonable vocabulary coverage, coherent noun context windows, and dense lexical chain connectivity are systematically violated.

The Baseline’s superior performance in our setting illustrates a broader principle: semantic enrichment is most valuable when raw term matching is insufficient. For formal news text with constrained vocabulary, TF-IDF leaves substantial semantic information uncaptured, and WSD plus lexical chains provide meaningful additional signal. For informal discussion text with diverse, topic specific jargon, TF-IDF already captures strong discriminating patterns through direct term frequency, and the semantic enrichment steps introduce more noise than signal.

Notably, the partial recovery from DC to DCS confirms that lexical chain filtering has a genuine noise reduction effect, even on informal text. With a more conservative filtering threshold, a larger corpus, or supplementary semantic resources, DCS performance might approach or match the Baseline. This partial improvement suggests that the pipeline’s architecture is sound, and that adaptation to informal corpora is a tractable problem rather than a fundamental barrier.

7.2 Limitations and Future Directions

Several directions merit future investigation. First, implementing the full hybrid similarity measure from the paper (incorporating gloss overlap in addition to structural Wu-Palmer distance) would provide a more complete test of the original methodology; our reproduction uses the structural measure alone, and the hybrid may achieve more accurate sense assignments. Second, domain adapted POS tagging trained on informal text — would improve noun extraction quality. Third, supplementing WordNet with broader semantic resources such as BabelNet or DBpedia would reduce the 20.2% coverage gap that disproportionately affects technical and informal categories. Finally, exploring adaptive chain score thresholds calibrated to corpus level chain density could prevent the over reduction observed in DCS, potentially bringing it closer to Baseline performance on informal text.

8. Conclusion

This report presents a reproduction of the semantic text clustering framework of Wei et al. (2015), applied to the 20 Newsgroups dataset. All three pipeline variants Baseline (TF-IDF), DC (WSD + TF-IDF), and DCS (WSD + Lexical Chains + TF-IDF) were implemented and evaluated on 600 balanced documents across six news categories.

The Baseline achieves the strongest performance, while both DC and DCS fall below it. This reversal of the trend observed in the original paper is primarily attributable to the informal, internet based nature of the 20 Newsgroups corpus, which reduces WordNet coverage, increases WSD error rates, and produces sparse lexical chain structures. Despite not replicating the paper’s quantitative improvements, our reproduction confirms that the pipeline behaves

coherently: DCS partially recovers from DC's shortfall through lexical chain filtering, and both semantic systems achieve meaningful dimensionality reduction.

The study highlights that the effectiveness of WordNet based semantic enrichment is closely tied to the formality and register of the input corpus, and that results on formal newswire text do not transfer directly to informal internet discussion. The pipeline's architecture is sound; successful application to informal corpora would require higher WordNet coverage, improved WSD accuracy, and more adaptive feature selection thresholds.